

Additive Manufacturing of High-Entropy Alloys for Marine Applications: Processing, Corrosion Performance, and Pathways to Qualification

Anirban Chakraborty, Pankaj Sharma*, and Chinmoy Sarkar
Indian Maritime University

Abstract—High-entropy alloys (HEAs) combine high strength, structural stability, and—in carefully chosen compositions—passive-film corrosion resistance that can exceed conventional marine alloys in chloride media. Additive manufacturing (AM) is a natural processing partner for these alloys: rapid solidification suppresses detrimental elemental segregation, refines the microstructure, and can improve pitting resistance relative to cast material, while powder- and wire-based routes enable complex wetted components, large structures, and in-service repair that are uneconomical by casting or forging. This review synthesizes the AM processing of HEAs (LPBF, SEBM, DED/LMD, WAAM, and binder jetting), the as-built and heat-treated microstructures that result, and their electrochemical and corrosion performance in saline and seawater environments, before mapping these capabilities onto concrete marine applications: seawater piping, pumps and valves, heat exchangers, propulsion components, and cladding-based repair of shafts and propellers. We show that the main scientific gaps are environment-specific: long-term natural seawater exposure, biofilm-coupled corrosion, corrosion fatigue and stress corrosion cracking data, and galvanic compatibility of HEA deposits on marine steels are all poorly quantified. The main engineering gaps are feedstock cost, defect control, and the absence of classification-society qualification pathways for HEA compositions. We close with a staged roadmap linking alloy design, process control, and standards-based qualification for marine deployment.

Index Terms—High-entropy alloys, additive manufacturing, laser powder bed fusion, wire arc additive manufacturing, directed energy deposition, seawater corrosion, pitting, passivation, marine engineering, qualification.

I. INTRODUCTION AND SCOPE

Corrosion in marine and offshore systems is a first-order economic and safety problem. The global cost of corrosion has been estimated at roughly USD 2.5 trillion per year, on the order of 3–4% of world gross domestic product, with marine infrastructure, shipping, and offshore energy accounting for a substantial share [1]. Seawater is among the most aggressive natural electrolytes: a near-neutral solution containing approximately 0.5–0.6 mol/L of chloride, dissolved oxygen, biological activity, and wide swings in temperature, flow velocity, and aeration between the atmospheric, splash, tidal, and fully immersed zones [2].

Conventional marine alloys each carry well-known limitations. Austenitic 316L stainless steel relies on a chromium-rich passive film but remains susceptible to pitting, crevice corrosion, and stress corrosion cracking (SCC) in warm, stagnant, or

creviced chloride service [3]. Nickel–aluminum bronzes resist biofouling reasonably well but suffer dealuminification and cavitation damage; copper–nickel alloys trade strength for antifouling behavior; duplex stainless steels and titanium alloys perform well but are expensive and difficult to fabricate into complex or large geometries [2]. A material family that combines strength, toughness, and robust passivity in chloride media—and that can be formed into complex, repairable components—would therefore address a genuine gap.

High-entropy alloys (HEAs), defined by five or more principal elements in near-equi-molar or deliberately non-equi-molar ratios [4], [5], are credible candidates. Their high configurational entropy, severe lattice distortion, sluggish diffusion, and the so-called “cocktail” effect together stabilize simple solid-solution phases and, in Cr-bearing compositions, support protective passive films [6]–[8]. Reviews of HEA corrosion behavior identify compositions—particularly FCC CoCrFeMnNi-type and carefully balanced Al-containing eutectic alloys—whose pitting resistance in chloride solutions rivals or exceeds 316L [9], [10].

Additive manufacturing (AM) completes the picture for three reasons. *First*, AM’s rapid solidification ($\sim 10^3$ – 10^6 K/s in powder-bed processes) suppresses the coarse segregation typical of cast HEAs, and recent evidence shows that this homogeneity can translate directly into better chloride corrosion performance [11], [12]. *Second*, AM enables geometries that casting cannot deliver economically: conformally cooled heat exchangers, topologically optimized brackets, and integrated manifolds for seawater systems [13]. *Third*, directed energy deposition (DED) and wire arc additive manufacturing (WAAM) make HEAs practical as repair and cladding materials for large marine components, avoiding the need to build entire structures from expensive alloys [14], [15].

Existing reviews have treated AM of HEAs comprehensively from the processing and mechanical-property viewpoints [12], [13], [16], and marine-corrosion texts mention HEAs as an emerging material class [2], but no synthesis to date is organized around *marine deployment*: which compositions survive real seawater service, which AM routes suit which marine components, and what qualification evidence is missing. This review targets that gap. Its scope is: (i) HEA fundamentals relevant to marine service (Section II); (ii) AM processes and feedstock for HEAs (Section III); (iii) microstructure and mechanical performance of AM HEAs (Section IV); (iv) corrosion and electrochemical behavior in saline and seawater

*Corresponding author. Email: pankajs@imu.ac.in

environments (Section V); (v) candidate marine applications and qualification pathways (Section VI); and (vi) challenges and a deployment roadmap (Sections VII–VIII). Table and figure sets are collected in Section IX.

II. HEA FUNDAMENTALS RELEVANT TO MARINE SERVICE

A. Core effects, critically stated

The original HEA concept invoked four “core effects”: high configurational entropy stabilizing solid solutions, severe lattice distortion impeding dislocation motion and diffusion, sluggish diffusion retarding phase decomposition, and a cocktail effect in which properties emerge non-linearly from the mixture [4], [8]. A critical reassessment by Miracle and Senkov [6] retains configurational entropy as a genuine thermodynamic contribution but cautions that it alone does not guarantee single-phase microstructures, that lattice distortion is composition-dependent, and that “sluggish diffusion” is neither universal nor necessarily large. For the marine engineer the practical restatement is simple: HEA compositions must be evaluated case by case, and phase stability in the presence of defects, segregation, and service temperature is what matters [7].

B. Phase families and their marine relevance

FCC alloys (prototypically CoCrFeMnNi, the “Cantor” alloy [5]) are ductile at all temperatures, with strength rising as temperature falls [17] and fracture toughness that remains exceptional even at cryogenic temperatures [18]. This combination is attractive for seawater piping, cryogenic cargo systems, and any component where corrosion resistance must coexist with toughness. Their weakness is moderate yield strength, which must be supplied by grain refinement, cold work, or precipitation engineering.

BCC and refractory HEAs (e.g., Nb-, Mo-, Ta-, W-rich systems) offer high strength and thermal stability but typically poorer ambient ductility and, depending on composition, weaker passivity; their AM processing is reviewed in [16]. For marine use they are best viewed as candidates for hot or wear-dominated niches rather than bulk seawater-wetted structures.

Eutectic HEAs (EHEAs), led by AlCoCrFeNi_{2.1} with its lamellar FCC/B2 duplex structure [19], balance strength and ductility and—importantly for AM—solidify as a coupled eutectic, which reduces hot cracking susceptibility. Recent work shows that the corrosion behavior of this family is exquisitely sensitive to how uniformly the elements partition between the two phases [11], making AM’s rapid solidification a genuine processing lever.

C. Passivity and its limits in chloride media

HEA corrosion resistance is not automatic. The most influential corrosion review of HEAs, by Qiu et al. [9], established that Cr-rich passive films are the basis of good performance, while Al- and Ni-rich BCC/B2 phases in multiphase alloys are preferential pitting sites; later work on Al_xCoCrFeNiTi_y quantified how microstructural evolution shifts electrochemical response [10]. Two directions are directly relevant to seawater

service. First, *computational alloy design for passivity*: Lu et al. [20] demonstrated a thermodynamics-plus-electrochemistry design loop that produced a corrosion-resistant HEA for harsh environments, and Quiambao et al. [21] characterized its passivation in detail, showing that selective Cr and Fe enrichment of the passive film underlies its resistance. Second, *Mo and W additions* strengthen passive films in chloride solutions in much the same way Mo does in stainless steels, a strategy recently exploited in EHEAs (Section V).

Two marine-specific caveats follow. Passive-film stability is *environment-specific*: an alloy that passivates in dilute or neutral chloride can behave very differently in acidic, sulfide-bearing, or creviced conditions typical of stagnant seawater or under-biofilm niches. And the “cocktail effect” works both ways: beneficial synergies are documented, but adverse ones—e.g., micro-galvanic coupling between constituent phases—are equally possible [2], [9].

III. ADDITIVE MANUFACTURING ROUTES FOR HEAS

AM of HEAs has been reviewed comprehensively by Cagirici et al. [13] and Han et al. [12]; here we summarize the process–property consequences that matter for marine components, then discuss feedstock, which is the binding constraint for marine-scale deployment. Table I (Section IX) compares the routes.

A. Powder bed fusion: LPBF and SEBM

Laser powder bed fusion (LPBF) melts thin powder layers with a focused laser, producing the highest cooling rates ($\sim 10^5$ – 10^6 K/s) and the finest as-built microstructures of any practical HEA route: cellular and dendritic substructures with sub-micron dislocation cells and strong crystallographic texture along the build direction [12], [13]. The same rapid solidification that refines the structure is what suppresses elemental segregation in multiphase HEAs; this is the microstructural origin of the improved chloride corrosion resistance reported for LPBF CoCrFeMnNi and SLM AlCoCrFeNi_{2.1} relative to cast material [11], [22]. The penalties are limited build volume, slow build speed, and susceptibility to lack-of-fusion and keyhole porosity when parameters are off-window; high-Al compositions additionally face solidification cracking [13].

Selective electron beam melting (SEBM/EBAM-class processes) operates under vacuum with a heated powder bed, reducing thermal gradients and residual stress. This makes SEBM the preferred powder-bed route for crack-prone and refractory systems [16], and its preheat capability is valuable for BCC HEAs with ductile-to-brittle transitions. Surface finish and resolution are coarser than LPBF, so wetted surfaces intended for corrosion service require machining or polishing.

B. Directed energy deposition and laser metal deposition

DED/LMD feeds powder or wire into a moving melt pool. Its marine significance is twofold. First, it enables *cladding and repair*: HEA overlays on marine steel substrates or worn bronze/steel components deliver corrosion- and wear-resistant surfaces at a fraction of a monolithic HEA part’s cost; laser-clad AlCoCr_xFeNi coatings on steel have been studied specifically

in this light [14]. Second, DED supports *in-situ alloying and compositionally graded deposits*, which researchers have used both to screen compositions and to functionally grade from a steel substrate to a corrosion-resistant HEA surface [13]. Within this route, Ti- or Cu-doped AlCoCrFeNi deposits and Nb–Mo–Ta–W compositions designed for marine service have been reported; the former showed improved corrosion resistance, while the latter—despite better overall performance than 316L—exhibited localized pitting associated with minor compositional segregation, illustrating the defect sensitivity discussed in Section V [13]. Control of dilution and interfacial galvanic behavior is the key engineering issue for clad components.

C. Wire arc additive manufacturing

WAAM deposits material by arc welding of wire feedstock at deposition rates and build volumes far beyond powder-bed processes, which is why it is the route of record for large marine and offshore structures [15]. For HEAs the bottleneck has been wire availability, and three strategies have emerged: cable-type twisted wires (demonstrated for MoNbTaWTi, yielding single-phase BCC deposits with hardness near 533 HV and compressive strength stable between 500 and 1000 °C [23]), combined-cable wire for Al–Co–Cr–Fe–Ni [24], and metal powder-cored wires, now demonstrated for Cantor-type CoCrFeMnNi walls with homogeneous element distribution and competitive mechanical properties [25]. WAAM HEA deposits are coarse and columnar, with cyclic reheating from successive passes; Mn evaporation, oxide and sulfide inclusions, and mild elemental segregation (notably Mn) are recurring metallurgical issues [15], [25]. Because WAAM parts are typically machined to finish, the as-deposited corrosion performance is less critical than the properties of the final machined surface—but inclusions and lack-of-fusion defects remain potential pit initiation sites.

D. Binder jetting and other routes

Binder jetting followed by sintering offers high build speed and low cost but leaves residual porosity unless infiltration or hot isostatic pressing (HIP) is applied; porous binder-jet HEA structures have been proposed for lightweight or functional applications [13]. For load-bearing, seawater-wetted marine parts, residual porosity is a corrosion liability (Section V), so binder jetting is currently a secondary route for this application domain.

E. Feedstock: the binding constraint

Marine deployment is ultimately limited less by process physics than by feedstock economics. Gas-atomized pre-alloyed HEA powders are expensive, and the critical-raw-material content (Co, Ni, Ta, Nb) of many attractive compositions raises both cost and supply-risk questions [12], [13]. Wire production for WAAM—drawing, cabling, or powder-coring—is immature for most HEA compositions, and element evaporation (Mn, Cr) during wire manufacture and deposition must be compensated in the chemistry [25]. Powder reuse and oxidation control in the humid, saline atmosphere of shipyard environments add further qualification requirements. A credible marine HEA program therefore starts with a feedstock strategy, not a printer.

IV. AS-BUILT MICROSTRUCTURE AND MECHANICAL PERFORMANCE

A. Characteristic AM microstructures of HEAs

Across powder-bed processes, as-built HEAs display fine cellular or dendritic solidification substructures, high dislocation densities arranged in cell walls, and epitaxially grown columnar grains aligned with the build direction; texture, cell size, and phase balance are all tunable through laser power, scan speed, hatch spacing, and scan strategy [12], [13]. In single-phase FCC alloys such as CoCrFeMnNi, LPBF produces a homogeneous solid solution with grain sizes well below cast equivalents, which raises yield strength via the Hall–Petch effect while retaining ductility [22]. In eutectic AlCoCrFeNi_{2.1}, SLM produces a fine, relatively homogeneous FCC/B2 duplex structure whose inter-phase composition spread is far smaller than in cast or annealed material [11]. WAAM deposits are coarser and more anisotropic: elongated columnar grains, dendritic substructure, and reheated interpass regions [23], [25].

B. Defects and their engineering meaning

The defect population of AM HEAs mirrors that of AM metals generally: lack-of-fusion pores, gas porosity, solidification cracks, and inclusions, plus HEA-specific risks such as hot cracking in Al-rich compositions and elemental loss by evaporation [13]. For marine service the important point is that every pore or crack intersecting a wetted surface is a candidate pit or crevice initiation site, and internal porosity can connect into crevice-like networks after surface machining [2]. Hot isostatic pressing closes internal porosity and is now standard practice for qualifying powder-bed parts; HIPed LPBF CoCrFeMnNi retains essentially the same chloride corrosion performance as the as-built state while improving density [22].

C. Post-processing: the heat-treatment trade-off

Heat treatment of AM HEAs is a genuine design decision with mechanical benefits and corrosion costs. Homogenization anneals relieve residual stress and coarsen the microstructure, but they can also restore the elemental segregation that rapid solidification suppressed. The clearest evidence comes from SLM AlCoCrFeNi_{2.1}: the as-built alloy, with its homogeneous inter-phase chemistry, shows superior corrosion resistance in 3.5 wt% NaCl, while annealing dramatically degrades passivity as FCC/B2 segregation develops [11]. Similar homogenization–corrosion interactions are reported for Al_xCoCrFeNi systems [12]. Conversely, targeted annealing of LDED AlMo_{0.25}FeCoCrNi_{2.1} has been used to tune strength and corrosion jointly [26]. The design rule: any post-build heat treatment must be qualified for corrosion, not just mechanics.

D. Representative mechanical performance

A few results anchor the current state of the art. LPBF CoCrFeMnNi achieves strength above the wrought baseline with retained ductility, and exhibits notable abrasion resistance—the first low-stress abrasion data reported for any AM HEA [22]. Mo-alloyed EHEAs reach yield strengths near 926 MPa

with compressive strains approaching 40% [27]. WAAM MoNbTaWTi deposits remain hard (around 533 HV) and maintain compressive strength from 500 to 1000 °C, positioning them for hot, abrasive niches such as exhaust-side marine hardware [23]. WAAM Cantor-type walls deliver competitive tensile behavior with anisotropy tied to the columnar grain structure [25]. Finally, hydrogen permeation through LPBF CoCrFeMnNi is markedly lower than through 316L despite a higher diffusion coefficient, a favorable sign for hydrogen uptake and embrittlement resistance in cathodically protected seawater systems [22].

V. CORROSION PERFORMANCE IN SALINE AND SEAWATER ENVIRONMENTS

A. Test environments and methods

Almost all published AM-HEA corrosion data come from laboratory saline media—most commonly 3.5 wt% NaCl or artificial seawater prepared to ASTM D1141 [28]—using potentiodynamic polarization and electrochemical impedance spectroscopy under the conventions of ASTM G3/G5/G61 [29]–[31]. These tests rank alloys and passive-film quality well, but they are not seawater service: they omit biofilms, tidal wet–dry cycling, crevice geometries, flow, and multi-month timescales. Data from natural marine exposure exist for conventional AM alloys (e.g., SLM AlSi10Mg immersed in the Adriatic for 3–6 months [32]), but equivalent long-term exposures of AM HEAs are, to our knowledge, absent from the literature. This is the single most important evidence gap in the field.

B. AM versus cast: the homogeneity advantage

The consistent finding across recent studies is that AM improves the chloride corrosion performance of single-phase and near-single-phase HEAs relative to cast counterparts. Mechanisms are twofold: (i) homogeneous elemental distribution removes micro-galvanic couples between segregated phases, and (ii) lower micro-porosity and finer structure reduce pit initiation sites. Reported results include LPBF CoCrFeMnNi with higher pitting resistance and polarization resistance than cast material [12], and SLM CoCrFeMnNi whose corrosion current density drops by roughly 58% relative to the as-cast alloy, attributed to a more uniform composition and a more stable passive film [12], [13]. In AlCoCrFeNi_{2.1} EHEA, the as-built SLM condition outperforms both annealed and cast specimens in 3.5 wt% NaCl for exactly these reasons [11]. The caveat, again, is that these advantages are hostage to defect control: the same reviews flag that secondary phases, intermetallics, and segregation introduced by poor parameters or heat treatment can create localized corrosion sites [13].

C. AM HEAs against the marine benchmark: 316L

The most direct marine-relevant comparison is LPBF CoCrFeMnNi against LPBF 316L. In chloride solution, as-built and HIPed CoCrFeMnNi exhibit roughly 2.2 times better corrosion performance than 316L, with stable pseudo-passive behavior and higher pitting potentials. In acidic solution the ranking inverts completely: 316L outperforms the HEA by more than

an order of magnitude [22]. The lesson for marine specification is that HEA superiority in seawater is real but *environment-specific*; acidic, sulfide-rich, or creviced microenvironments (under deposits, inside crevices, beneath biofilms) cannot be assumed benign on the strength of neutral NaCl tests alone [9].

D. Composition strategies for seawater passivity

Several alloying strategies are emerging specifically for chloride service. Mo additions to EHEAs (AlCoCrFeNi_{2.1}Mo_x) simultaneously raise strength and pitting potential, with in-situ Mo-oxide/Cr-oxide composite films stabilizing passivity [27]. Ti or Cu doping of LMD AlCoCrFeNi improves corrosion resistance over the undoped alloy, though secondary intermetallics must be suppressed [13]. Zr-containing HEAs have shown improved resistance over Zr-free variants [13], and Nb–Mo–Ta–W refractory compositions designed for marine use beat 316L overall, albeit with deep pits nucleating at minor segregates in one study [13]. CoCrW-type alloys deposited by LPBF show passivation that evolves over weeks of immersion, with corrosion concentrated at melt-pool boundaries—a reminder that AM’s own solidification pattern becomes part of the corrosion microstructure [33].

E. Anisotropy, surface condition, and coupled degradation

Three second-order effects are known from AM metals generally and are only beginning to be characterized in HEAs. *Build orientation* changes the density of melt-pool boundaries and cell structures exposed at a surface, and corrosion rates in LPBF alloys vary with orientation [12], [13]. *Surface condition* dominates localized attack: for LPBF 316L, as-built rough surfaces are markedly more SCC-susceptible than machined ones, and cracks propagate along melt-pool boundaries [3]; machined AM surfaces can even exceed wrought behavior in pitting tests [2]. *Coupled degradation*—corrosion fatigue, SCC, erosion-corrosion, and cavitation—is where marine components actually fail, yet AM-HEA data in these modes are essentially nonexistent; hydrogen permeation evidence for LPBF CoCrFeMnNi is one of the few positive indicators [22].

F. Galvanic and biofouling interactions

For DED-clad or WAAM-on-steel components, galvanic coupling between the HEA deposit and the substrate (or between an AM HEA part and adjacent conventional alloys) must be assessed; passive HEAs are typically noble relative to structural steel, so coating defects concentrate attack on the substrate side, and compatibility with cathodic protection regimes (e.g., offshore CP design practice) needs verification [2]. Biofouling interactions with HEA surfaces—which couple biological films to under-film chemistry and may accelerate localized attack—have not yet been studied in any systematic way. Table II summarizes the key corrosion evidence base.

VI. MARINE APPLICATIONS AND DEPLOYMENT PATHWAYS

A. Where AM HEAs fit in the marine system

The argument for AM HEAs is strongest where corrosion cost, geometric complexity, or repair difficulty are simultaneously high, so that the material premium is amortized. Table III maps candidate applications; the main clusters are:

Seawater systems and pumps. Impellers, valve trim, strainers, and fittings in ballast, cooling, and fire-main systems are small, complex, and corrosion-critical—an ideal LPBF niche, echoing LPBF’s established role in complex high-performance components [13]. FCC CoCrFeMnNi-type alloys and Mo-modified EHEAs are the leading composition candidates given their chloride performance [22], [27].

Heat exchangers and desalination hardware. LPBF’s ability to print conformal internal channels enables compact seawater heat exchangers; the same geometrical freedom applies to energy-recovery and reverse-osmosis components, where pitting and crevice resistance govern life [2], [13].

Large structures and propulsion. WAAM suits propeller blanks, brackets, offshore nodal repairs, and other meter-scale items where powder-bed volumes are impractical [15]; the economics depend on HEA wire becoming available at scale [25].

Repair and cladding. DED overlays of HEA on marine steel, or on worn bronze and steel components (shafts, seal seats, rudder stocks), deliver corrosion- and wear-resistant surfaces only where needed [14]. This is arguably the fastest route to impact because it sidesteps whole-part cost and leverages existing shipyard welding infrastructure.

B. Hybrid and functionally graded strategies

Because feedstock cost dominates, the near-term deployment model is hybrid: conventional marine steel for structure, AM HEA where the seawater actually acts. Compositionally graded DED builds—steel to HEA transition—can manage both cost and galvanic compatibility, provided dilution and intermetallic formation at the interface are controlled [13]. Cathodic protection compatibility must be checked case by case, since passive HEAs shift the galvanic balance of a protected assembly [2].

C. Qualification: what classification societies need

Marine deployment requires classification-society acceptance. DNV’s standard for additive manufacturing of metallic parts (DNV-ST-B203 [34]), ABS’s metal AM guidelines [35], and Lloyd’s Register’s AM requirements [36] currently provide frameworks developed around conventional engineering alloys; HEA compositions sit outside the qualified material list and would need to be qualified as new materials, with mechanical and corrosion datasets. A defensible qualification ladder for an AM HEA marine component is:

- 1) *Material baseline:* composition control, powder/wire certification, defect acceptance criteria (porosity, cracks, inclusions) with NDT.
- 2) *Electrochemical screening:* polarization and EIS in ASTM D1141 seawater [28] under ASTM G3/G5/G61

conventions [29]–[31], benchmarked against the incumbent alloy (e.g., 316L, NiAl bronze) in identical conditions.

- 3) *Long-term exposure:* immersion and wet–dry cycling in natural or artificial seawater for months, with post-exposure mechanical re-testing, mirroring exposure protocols already used for conventional AM alloys [32].
- 4) *Coupled degradation:* corrosion fatigue and SCC (e.g., slow-strain or U-bend tests in seawater), erosion-corrosion and cavitation for rotating parts [2], [3].
- 5) *System level:* galvanic compatibility tests in the actual assembly, CP compatibility, and pilot installation with inspection intervals.

Table IV maps the applicable standards onto this ladder. The gap worth emphasizing: no existing marine AM standard references HEA compositions, and no HEA standard references marine exposure classes—closing that two-way gap is the qualification task.

VII. CHALLENGES AND OUTLOOK

A. Open technical challenges

Feedstock economics and supply. Powder costs and critical-element exposure (Co, Ni, Ta, Nb) remain the largest barrier to marine adoption; WAAM wire routes mitigate cost but introduce evaporation and inclusion control problems [13], [25]. Development of low-Co or Co-free corrosion-resistant HEAs is strategically important.

Defect–corrosion linkage. The field lacks quantitative defect-tolerance criteria: how much porosity, of what size and connectivity, can a seawater-wetted AM HEA surface tolerate before pitting statistics become unacceptable? Such criteria exist implicitly for aerospace fatigue but not for marine corrosion, and they will be needed for classification [34].

Environment-specificity of passivity. The 316L-comparison evidence shows HEA passivity can collapse outside neutral chloride media [22]. Crevice chemistry, under-biofilm acidification, and sulfides in stagnant zones all push toward those regimes and are untested for AM HEAs [9].

Coupled degradation modes. Corrosion fatigue, SCC, cavitation, and erosion-corrosion data for AM HEAs are nearly absent, yet these govern marine component life [3].

Biofouling–corrosion interaction. Whether HEA surfaces, with their distinct passive-film chemistry, resist, tolerate, or accelerate biofilm-mediated attack is simply unknown; this connects the present topic to the companion literature on sustainable biofouling control [2].

B. Research directions

Corrosion-by-design. The demonstrated loop of thermodynamic design plus electrochemical validation [20], [21] should be closed with machine-learning surrogate models trained on AM-specific microstructures, targeting compositions that are simultaneously printable (hot-crack resistant) and passive in seawater.

Process–passivity maps. Systematic studies linking LPBF/WAAM parameter windows to passive-film composition (e.g., Cr/Mo enrichment measured by surface analysis)

would convert corrosion from a screening outcome into a process-controlled property [11], [33].

Real-seawater exposure networks. Long-term immersion programs—tidal, splash, and submerged zones, with and without biofilm—for a small matrix of benchmark AM HEAs versus incumbent alloys, following the natural-exposure template already demonstrated for AM aluminum [32].

Hybrid structures. Functionally graded DED builds and HEA claddings on marine steel, with galvanic and CP compatibility designed in, are the most economical near-term pathway [13], [14].

Digital qualification. In-situ melt-pool monitoring plus build-data archives could support the data-driven material qualification frameworks classification societies are moving toward [34], [35], shortening the path for each new HEA composition.

C. Deployment roadmap

A realistic staged roadmap is: (1) *now*—benchmarking studies and repair cladding pilots on non-class-critical hardware; (2) *near term* (2–5 years)—qualified LPBF small components in seawater systems, WAAM propeller and bracket demonstrations, and the first natural-seawater exposure datasets; (3) *medium term* (5–10 years)—class-approved HEA compositions within marine AM standards, functionally graded marine structures, and life-cycle-cost validated substitution of incumbent alloys. Sustainability assessment should run alongside: HEA-AM parts must justify their critical-element burden through longer life, less maintenance, and recyclability to be credible in an industry moving toward life-cycle accountability [2].

VIII. CONCLUSIONS

- 1) HEAs with FCC CoCrFeMnNi-type or Mo-modified eutectic compositions exhibit passive-film corrosion resistance in neutral chloride media that matches or exceeds the 316L benchmark, making them credible candidates for seawater-wetted marine components [9], [22], [27].
- 2) Additive manufacturing is not merely a shaping tool but a *corrosion-engineering* tool for HEAs: rapid solidification suppresses elemental segregation, refines microstructure, and measurably improves pitting resistance over cast material—provided defects and post-build segregation are controlled [11]–[13].
- 3) Process selection maps onto marine needs: LPBF for complex small components in seawater systems, SEBM for crack-prone and refractory systems, DED for cladding, repair, and functionally graded construction, and WAAM for large structures once HEA wire feedstock matures [13]–[15].
- 4) Corrosion performance is environment-specific: AM CoCrFeMnNi surpasses 316L in chloride but is far inferior in acidic media, so crevice, under-deposit, and biofilm microenvironments must be tested explicitly rather than extrapolated from NaCl screening [22].
- 5) The decisive gaps for deployment are evidential and regulatory: long-term natural seawater exposure, corrosion fatigue/SCC, biofilm coupling, and galvanic/CP

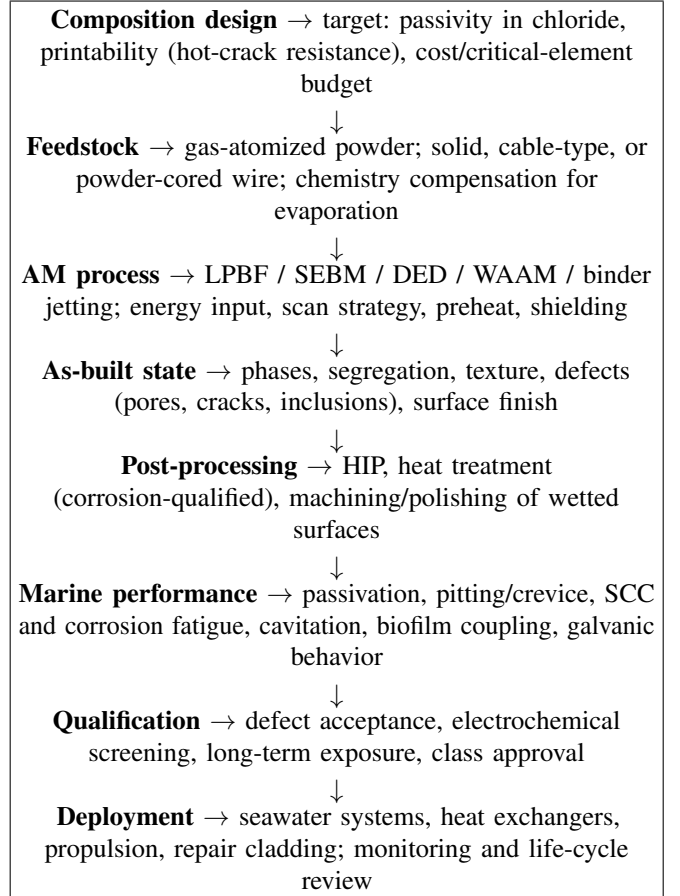


Fig. 1. Process–structure–property–qualification chain for AM HEAs in marine service. The framework organizes the review: each stage is a control point, and the weakest link (long-term seawater evidence and standards coverage) defines the deployment roadmap.

compatibility data are scarce, and no marine AM standard yet accommodates HEA compositions [34]–[36].

- 6) The practical near-term pathway is hybrid: HEA claddings and graded deposits on marine steel, qualified through a staged ladder of electrochemical screening, long-term exposure, coupled-degradation testing, and system-level galvanic assessment.

IX. TABLES AND FIGURES

REFERENCES

- [1] G. H. Koch, M. P. H. Brongers, N. G. Thompson, Y. P. Virmani, and J. H. Payer, “International measures of prevention, application, and economics of corrosion technologies (IMPACT),” NACE International, Tech. Rep., 2016.
- [2] “Sustainable marine structures,” 2025, nASS Publishing, chapter on marine corrosion protection materials. [Online]. Available: <https://journals.nasspublishing.com/index.php/sms/article/download/2335/1113/9931>
- [3] “Stress corrosion cracking of 316L manufactured by laser powder bed fusion in 6% ferric chloride solution,” 2022, corrosion Science (article record S0010938X2200453X). [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0010938X2200453X>
- [4] J.-W. Yeh, S. K. Chen, S. J. Lin, S. Y. Chin, T. T. Shun, C. H. Tsau, T. G. Chou, and S. Y. Chang, “Nanostructured high-entropy alloys with multiple alloying elements,” *Advanced Engineering Materials*, vol. 6, no. 6, pp. 299–303, 2004.
- [5] B. Cantor, I. T. H. Chang, P. Knight, and A. J. B. Vincent, “Microstructural development in equiatomic multicomponent alloys,” *Materials Science and Engineering A*, vol. 375–377, pp. 213–218, 2004.

TABLE I
ADDITIVE MANUFACTURING ROUTES FOR HEAS AND THEIR MARINE RELEVANCE (SYNTHESIZED FROM [12], [13], [15], [16]).

Process	Solidification and microstructure	Principal advantages	Principal limitations	Marine relevance
LPBF (SLM)	Very high cooling rate ($\sim 10^5$ – 10^6 K/s); fine cells, high dislocation density, build-direction texture	High resolution, complex geometry, best as-built homogeneity; fine parts	Small build volume, slow rate, porosity/cracking off-window, powder cost	Impellers, valves, fittings, heat-exchanger cores; complex seawater-system parts
SEBM	High cooling rate under vacuum; bed preheat; slightly coarser than LPBF	Low residual stress, good for crack-prone and refractory HEAs	Coarser surface, equipment cost, limited material library	Crack-sensitive or refractory compositions; parts later machined to wetted finish
DED / LMD	Moderate-to-high cooling; variable microstructure along build; in-situ alloying	Repair and cladding, graded deposits, high deposition, build on existing parts	Lower accuracy, dilution control, surface finish	Overlays on shafts, seal seats, rudder stocks; steel-to-HEA graded construction
WAAM	Low-to-moderate cooling; coarse columnar grains; interpass reheating	Meter-scale builds, high rate, low equipment cost, shipyard-compatible	Wire feedstock immaturity, Mn evaporation, inclusions, anisotropy	Propeller blanks, brackets, off-shore nodes, large repairs (machined to finish)
Binder jetting	No melting; sintered/infiltrated microstructure	Fast, cheap, batch production	Residual porosity unless infiltrated/HIPed; lower strength	Secondary route: porous/functional parts; limited for wetted load-bearing service

TABLE II
SELECTED CORROSION EVIDENCE FOR AM HEAS IN CHLORIDE MEDIA. ALL ENTRIES ARE LABORATORY SALINE TESTS UNLESS NOTED; LONG-TERM NATURAL SEAWATER DATA FOR AM HEAS REMAIN UNAVAILABLE.

Alloy	AM route	Medium	Key finding
CoCrFeMnNi	LPBF	3.5% NaCl	Higher pitting resistance and polarization resistance than cast; $\sim 58\%$ lower
CoCrFeMnNi	LPBF (as-built, HIPed)	3.5% NaCl vs. H_2SO_4	$\sim 2.2\times$ better than 316L in chloride; $\sim 15.5\times$ worse in acid; low H permea
AlCoCrFeNi _{2.1}	SLM	3.5% NaCl	As-built homogeneous duplex outperforms annealed and cast; annealing res
AlCoCrFeNi _{2.1} Mo _x	(cast; AM-adjacent design)	3.5% NaCl	Mo ($x=0.3$) raises strength and pitting potential via Mo/Cr composite oxid
AlMo _{0.25} FeCoCrNi _{2.1}	LDDED	Saline	Annealing heat treatment jointly tunes strength and corrosion response
AlCoCrFeNi (Ti, Cu doped)	LMD	Saline	Doping improves corrosion resistance over undoped; intermetallics must be
Nb–Mo–Ta–W	LPBF	Saline	Better overall than 316L; segregation sites produce pits up to ~ 1 mm deep
CoCrW	LPBF	Acidic NaCl, 28 d immersion	Oxide film matures then destabilizes; attack at melt-pool boundaries

- [6] D. B. Miracle and O. N. Senkov, "A critical review of high entropy alloys and related concepts," *Acta Materialia*, vol. 122, pp. 448–511, 2017.
- [7] E. P. George, D. Raabe, and R. O. Ritchie, "High-entropy alloys," *Nature Reviews Materials*, vol. 4, pp. 515–534, 2019.
- [8] B. S. Murty, J.-W. Yeh, S. Ranganathan, and P. P. Bhattacharjee, *High-Entropy Alloys*. Amsterdam: Elsevier, 2019.
- [9] Y. Qiu, S. Thomas, M. A. Gibson, H. L. Fraser, and N. Birbilis, "Corrosion of high entropy alloys," *npj Materials Degradation*, vol. 1, p. 15, 2017.
- [10] Y. Qiu, S. Thomas, D. Fabijanic, A. J. Barlow, H. L. Fraser, and N. Birbilis, "Microstructural evolution, electrochemical and corrosion properties of Al₂CoCrFeNiTi_y high entropy alloys," *Materials & Design*, vol. 170, p. 107698, 2019.
- [11] "Homogenization of AlCoCrFeNi_{2.1} eutectic high entropy alloy with improved corrosion resistance fabricated by selective laser melting," 2025, corrosion Science (article record S0010938X25004688). [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0010938X25004688>
- [12] F. Han, C. Li, J. Huang, J. Wang, L. Xue, C. Wang, and Y. Zhang, "Research advances in additively manufactured high-entropy alloys: Microstructure, mechanical properties, and corrosion resistance," *Metals*, vol. 15, no. 2, p. 136, 2025.
- [13] M. Cagirici, S. Guo, J. Ding, U. Ramamurty, and P. Wang, "Additive manufacturing of high-entropy alloys: Current status and challenges," *Smart Materials in Manufacturing*, vol. 2, p. 100058, 2024.
- [14] Y. Q. Jiang, J. Li, Y. F. Juan, Z. J. Lu, and W. L. Jia, "Evolution in microstructure and corrosion behavior of AlCoCr_xFeNi high-entropy alloy coatings fabricated by laser cladding," *Journal of Alloys and Compounds*, vol. 775, pp. 1–14, 2019.
- [15] L. P. Raut and R. V. Taiwade, "Wire arc additive manufacturing: A comprehensive review and research directions," *Journal of Materials Engineering and Performance*, vol. 30, no. 7, pp. 4768–4791, 2021.
- [16] M. R. Abdullah and Z. Peng, "Review and perspective on additive manufacturing of refractory high entropy alloys," *Materials Today Advances*, vol. 22, p. 100497, 2024.
- [17] F. Otto, A. Dlouhý, C. Somsen, H. Bei, G. Eggeler, and E. P. George, "The influences of temperature and microstructure on the tensile properties of a CoCrFeMnNi high-entropy alloy," *Acta Materialia*, vol. 61, no. 15, pp. 5743–5755, 2013.
- [18] B. Gludovatz, A. Hohenwarter, D. Catoor, E. H. Chang, E. P. George, and R. O. Ritchie, "A fracture-resistant high-entropy alloy for cryogenic applications," *Science*, vol. 345, no. 6201, pp. 1153–1158, 2014.
- [19] Y. P. Lu, Y. Dong, S. Guo, L. Jiang, H. J. Kang, T. M. Wang, B. Wen, Z. J. Wang, Y. P. Jie, Z. Q. Cao, H. H. Ruan, and T. J. Li, "A promising new class of high-temperature alloys: eutectic high-entropy alloys," *Scientific Reports*, vol. 4, p. 6200, 2014.
- [20] P. Lu, J. Saal, G. Olson, T. Li, O. Swanson, G. Frankel, A. Gerard, K. Quiambao, and J. Scully, "Computational materials design of a corrosion resistant high entropy alloy for harsh environments," *Scripta Materialia*, vol. 153, pp. 19–22, 2018.
- [21] K. Quiambao, S. McDonnell, D. Schreiber, A. Gerard, K. Freedy, P. Lu, J. Saal, G. Frankel, and J. Scully, "Passivation of a corrosion resistant high entropy alloy in non-oxidizing sulfate solutions," *Acta Materialia*, vol. 164, pp. 362–376, 2019.
- [22] "Laser powder bed fusion printing of CoCrFeMnNi high entropy

TABLE III
CANDIDATE MARINE APPLICATIONS FOR AM HEAS, PREFERRED ROUTES, AND ENABLING RATIONALE.

Marine subsystem	Candidate components	Preferred AM route	Enabling rationale	Readiness
Seawater systems	Impellers, valve trim, strainers, fittings	LPBF	Complex geometry; chloride passivity of FCC/EHEA compositions [13], [22]	Lab/pilot
Heat exchangers, desalination	Conformal-channel exchangers, RO hardware	LPBF	Internal-channel freedom; pitting/crevice resistance governs life [13]	Lab/pilot
Propulsion	Propellers, shafting features	WAAM (+ machining)	Meter-scale deposition; shipyard-compatible [15]	Lab/demonstrator
Structure and outfitting	Brackets, nodal repairs	WAAM / DED	Large builds; cost-effective deposition [15]	Lab/demonstrator
Repair and maintenance	Shaft, seal-seat, rudder-stock overlays	DED / LMD	Localized corrosion/wear resistance; graded interfaces [14]	Pilot (steel substrates)
Hot/abrasive niches	Exhaust-side hardware	SEBM, WAAM (RHEA)	High-temperature strength of refractory HEAs [16], [23]	Lab

TABLE IV
STANDARDS AND RULES FRAMEWORK RELEVANT TO QUALIFYING AM HEA COMPONENTS FOR MARINE SERVICE.

Document	Scope	Role in AM-HEA marine qualification
ASTM D1141 [28]	Preparation of substitute ocean water	Standard medium for reproducible seawater corrosion tests
ASTM G3, G5, G61 [29]–[31]	Electrochemical measurement conventions, potentiodynamic polarization, cyclic polarization	Screening and ranking of passive-film quality and pitting susceptibility
DNV-ST-B203 [34]	Additive manufacturing of metallic parts (marine)	Qualification framework to extend toward HEA compositions and defect-acceptance criteria
ABS metal AM guidelines [35]	Class approval of AM metal parts	Parallel route for machinery and hull components
Lloyd's Register AM requirements [36]	Class approval of AM processes and parts	Parallel route; data-driven qualification expectations

alloy: Processing, microstructure, and mechanical properties,” 2024, springer Link, DOI: 10.1007/s44210-024-00035-1. [Online]. Available: <https://link.springer.com/article/10.1007/s44210-024-00035-1>

- [23] J. Liu, J. Li, X. Du, Y. Tong, R. Wang, D. He, Z. Cai, and H. Wang, “Microstructure and mechanical properties of wire arc additively manufactured MoNbTaWTi high entropy alloys,” *Materials*, vol. 14, no. 16, p. 4512, 2021.
- [24] Q. Shen, X. Kong, and X. Chen, “Fabrication of bulk Al-Co-Cr-Fe-Ni high-entropy alloy using combined cable wire arc additive manufacturing (CCW-AAM): Microstructure and mechanical properties,” *Journal of Materials Science & Technology*, vol. 74, pp. 136–142, 2021.
- [25] R. Andersson, A. Zavdoveev, M. Speicher *et al.*, “Microstructural and mechanical characterisation of Cantor type high entropy alloy produced by wire arc additive manufacturing,” *The International Journal of Advanced Manufacturing Technology*, 2026.
- [26] “Microstructure, mechanical properties and corrosion resistance of laser direct energy deposited AlMo_{0.25}FeCoCrNi_{2.1} high-entropy alloy: adjusted by annealing heat treatment,” 2024, virtual and Physical Prototyping, DOI: 10.1080/17452759.2024.2385569. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/17452759.2024.2385569>
- [27] “The enhanced corrosion resistance and mechanical properties of AlCoCrFeNi_{2.1} eutectic high entropy alloy by addition of Mo element,” 2026, journal of Materials Research and Technology (article record S2238785426002796). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2238785426002796>
- [28] “ASTM D1141: Standard practice for the preparation of substitute ocean water,” aSTM International, West Conshohocken, PA.
- [29] “ASTM G3: Standard practice for conventions applicable to electrochemical measurements in corrosion testing,” aSTM International, West Conshohocken, PA.
- [30] “ASTM G5: Standard reference test method for making potentiodynamic anodic polarization measurements,” aSTM International, West Conshohocken, PA.
- [31] “ASTM G61: Standard test method for conducting cyclic potentiodynamic polarization measurements for localized corrosion susceptibility of iron-, nickel-, or cobalt-based alloys,” aSTM International, West Conshohocken, PA.
- [32] “Natural marine environment effect on additively manufactured aluminum alloy,” 2026, scienceDirect record S2452321626002799.

[Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2452321626002799>

- [33] “Corrosion behavior and mechanism of laser powder bed fusion produced CoCrW in an acidic NaCl solution,” 2023, corrosion Science (article record S0010938X23000410). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010938X23000410>
- [34] “DNV-ST-B203: Additive manufacturing of metallic parts,” dNV, Høvik, Norway.
- [35] “ABS guidelines for metal additive manufacturing,” american Bureau of Shipping, Houston, TX.
- [36] “Requirements for additive manufacturing,” Lloyd’s Register, London, UK.